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CyberShield: AI Based Smart Network Security System

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Abstract: In the modern cybersecurity landscape, small-scale networks often lack the resources for enterprise-grade hardware firewalls. This paper presents CyberShield, an Intrusion Prevention System (IPS) designed to run on consumer-grade hardware. By utilizing an Atheros AR9271 adapter to create a controlled wireless access point on an Ubuntu-based host, CyberShield intercepts and scores all transit traffic in real-time. Unlike traditional systems that rely on complex database management systems, CyberShield implements a high-efficiency logging mechanism using CSV-based flat files, integrated with a Streamlit-based Security Operations Center (SOC) dashboard. The result is a portable, effective defense system capable of automated DNS-based quarantine.

I. INTRODUCTION

In an era defined by the rapid expansion of the Internet of Things (IoT) and the decentralization of network infrastructures, edge-level security has become a critical frontline defence. As cyber threats evolve from simple malware to sophisticated, real-time anomalies, traditional security measures often struggle to keep pace without significant hardware investment. For small-scale environments and home networks, there is a clear and urgent need for a lightweight, responsive system that can not only monitor traffic but also take proactive measures to mitigate threats before they penetrate deeper into the network.

To address these vulnerabilities, this paper introduces CyberShield, an innovative Intrusion Prevention System (IPS) designed to provide enterprise-grade visibility on consumer-grade hardware. Unlike passive monitoring tools that simply alert a user to an intrusion after the fact, CyberShield is built as an active gateway. By positioning itself at the entry point of the network, the system acts as a "Defensive Man-in-the-Middle," ensuring that every packet transmitted across the wireless spectrum is subjected to rigorous real-time inspection and scoring.

The technical foundation of CyberShield relies on the strategic integration of specialized hardware and open-source software. By utilizing the Atheros AR9271 wireless adapter—renowned for its stability in Master (AP) mode—the system creates a dedicated `ap0` interface within an Ubuntu Linux environment. This setup allows the host machine to function as a wireless hotspot, forcing all connected client traffic through a centralized Python-based engine. This hardware-centric approach ensures that the system maintains high fidelity in packet capture without the packet loss often associated with purely software-based solutions.

A key innovation of the CyberShield architecture is its departure from heavy, resource-intensive database management systems. Instead of utilizing SQL-based backends, the project implements a high-efficiency Flat-File Database model using `Traffic log.csv` and `Security Log.csv`. This design choice minimizes CPU overhead and memory consumption, making the system ideal for low-power edge computing. This data is then consumed by a real-time Streamlit dashboard, providing administrators with a sophisticated, SOC-style visualization of network health, protocol distribution, and active threat alerts.

II. KEY COMPONENTS INCLUDE:

- The Atheros AR9271 Wireless Gateway acts as the primary physical entry point, utilizing its native support for Master Mode to establish the ap0 interface on an Ubuntu host.

- The In-Line Packet Sniffing Engine continuously monitors this interface in promiscuous mode, intercepting every raw frame that passes between the connected clients and the main internet connection.

- The Statistical Threat Scoring Logic processes the metadata of these intercepted packets in real-time, calculating a risk value based on traffic patterns and known anomaly signatures.

- The Flat-File Data Persistence Layer bypasses traditional database overhead by instantly appending all network telemetry and incident reports into Traffic log.csv and Security Log.csv.

- The Automated DNS Quarantine System serves as the mitigation layer, which, upon detecting a high threat score, triggers a script to isolate the offending device from the wider network.

- The Streamlit-Based SOC Dashboard provides the final visualization, dynamically reading the CSV logs to present a live security status and traffic analytics to the network administrator.

III. MOTIVATION OF THIS PROJECT

The widespread adoption of IoT devices and high-speed home networking has outpaced the security capabilities of standard consumer routers, leaving a significant vulnerability at the network edge. Most household and small-business users rely on basic firewalls that are incapable of deep packet inspection or real-time anomaly detection, making them easy targets for automated botnets and lateral movement attacks. The motivation for CyberShield stems from this security gap, aiming to provide a sophisticated yet accessible defence mechanism that can detect and stop threats at the point of entry before they compromise sensitive personal or corporate data.

Current enterprise-grade security solutions, while effective, are often prohibitively expensive and require significant computational resources, including dedicated servers and complex database management systems. This creates a barrier to entry for individual researchers and small organizations who need robust protection but lack the infrastructure to support heavy SQL-based Intrusion Prevention Systems. CyberShield is motivated by the need for "Democratic Security," proving that by using optimized Python logic and lightweight CSV-based data logging, it is possible to achieve high-performance threat detection on affordable, low-power hardware like an Ubuntu-powered laptop or a micro-computer.

Beyond mere detection, the project is driven by the necessity for automated, autonomous response in an era where cyberattacks happen at machine speed. Manual intervention is no longer a viable strategy for stopping modern network intrusions, which can exfiltrate data in seconds. The motivation behind integrating a hardware-based gateway with an automated DNS quarantine is to move away from passive logging and toward active defence. By building a system that can physically isolate a threat the moment it is logged, CyberShield empowers network administrators with a self-healing infrastructure that minimizes human error and significantly reduces the window of opportunity for attackers.

IV. PROBLEM STATEMENT

Modern smart homes use multiple IoT devices that remain always connected to the internet – but users have no visibility or control over what those devices are doing.

V. TECHNOLOGY

1. Operating System: Ubuntu Linux (Kernel 6.x)
Ubuntu serves as the backbone of the project because of its robust networking stack and extensive support for security tools. The Kernel 6.x series is specifically important as it includes updated drivers for modern hardware and enhanced support for nl80211, the standard used to manage wireless devices in Linux. It provides the stable environment necessary for low-level system calls and high-speed packet forwarding.

2. Hardware Capture: Atheros AR9271 (Master Mode)
The Atheros AR9271 chipset is the "gold standard" for wireless security research. Its primary role is to run in Master Mode, which allows the USB adapter to act as a physical Access Point (AP). Unlike standard Wi-Fi cards that only "receive" data, this hardware creates the ap0 interface, ensuring that all client traffic is physically forced through your Ubuntu host for inspection.

3. Development: Python 3.x
Python was selected as the primary language due to its rapid prototyping capabilities and its vast ecosystem of security libraries. In a research context, Python allows you to handle complex logic—like threat scoring and data visualization—with minimal code, making the system easier to maintain, debug, and explain in an academic paper.

4. Packet Processing: Scapy / Raw Sockets
Scapy is a powerful Python library used for packet manipulation and sniffing. It allows the CyberShield engine to perform Deep Packet Header Inspection. Instead of seeing traffic as a single "blob" of data, Scapy deconstructs it into layers (Ethernet, IP, TCP, UDP). This allows the engine to read the metadata (like source/destination and port numbers) to identify suspicious patterns.

5. Database: Flat-File (CSV)
Rather than using a complex database like SQL, you chose a Flat-File (CSV) system. This is a "performance-first" design choice. Appending a line to a .csv file requires almost zero CPU power compared to a SQL database, which has to manage indexes and

memory. This ensures the CyberShield engine stays fast and lightweight, preventing the logging process from slowing down the actual network speed.

6. Frontend: Streamlit

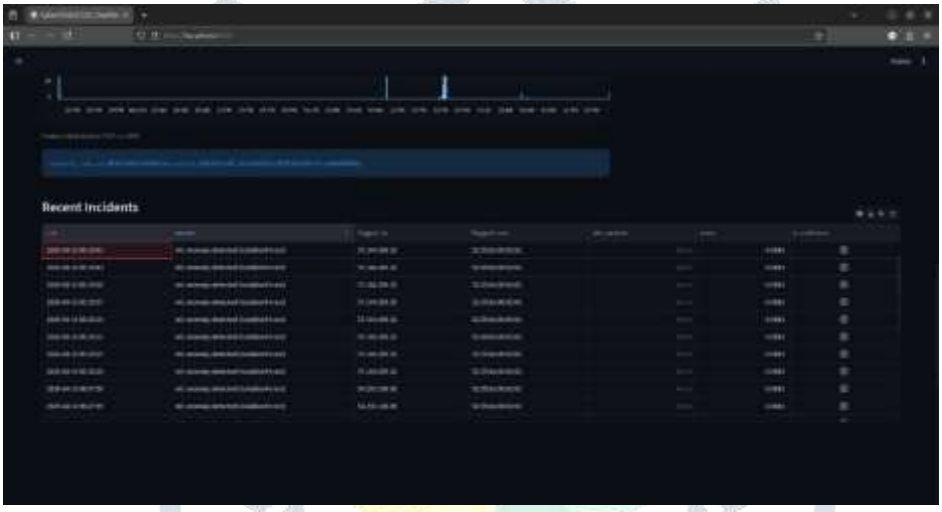
Streamlit is a modern framework used to turn data scripts into interactive web apps. It serves as your SOC (Security Operations Center) Dashboard. Streamlit is unique because it can "watch" your CSV files; the moment a new line is added to Traffic log.csv or Security Log.csv, Streamlit automatically refreshes the charts and alerts on your screen without you having to reload the page.

7. Mitigation: iptables / DNS Redirection

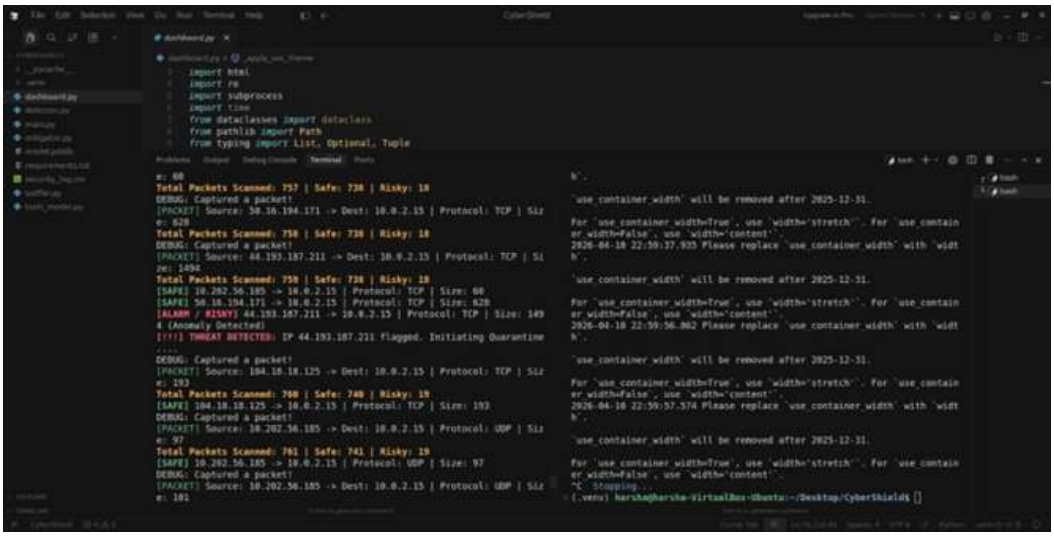
This is the "Defense" part of your Intrusion Prevention System:

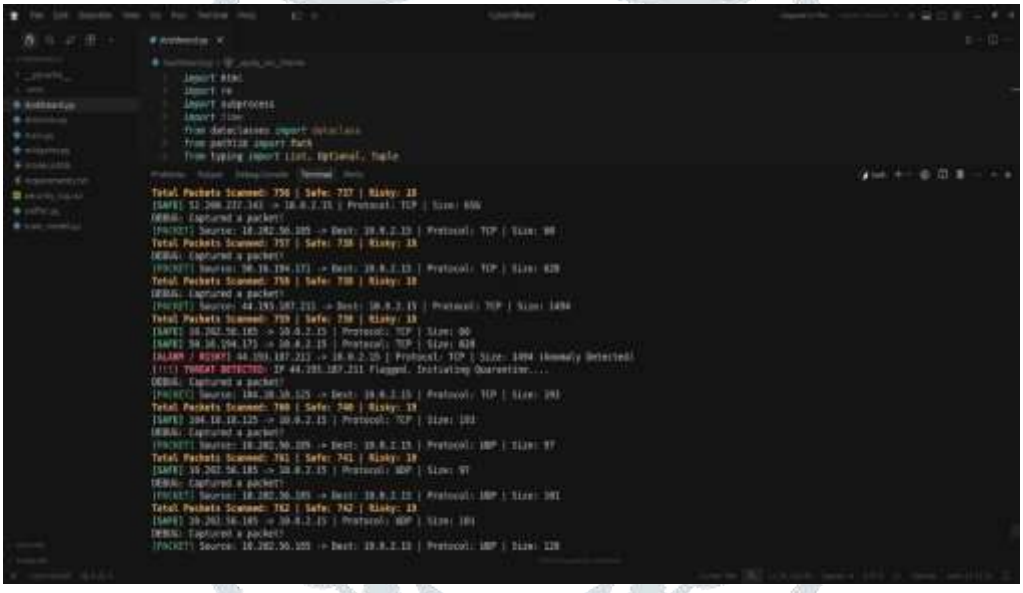
- iptables: This is the standard Linux firewall. When a threat is detected, CyberShield sends a command to iptables to block that specific device's MAC or IP address.
- DNS Redirection: Instead of just cutting the internet, the system redirects the user's web requests. If a compromised device tries to go to "google.com," the system intercepts the request and sends them to a local "Quarantine Page" that explains why they were disconnected.

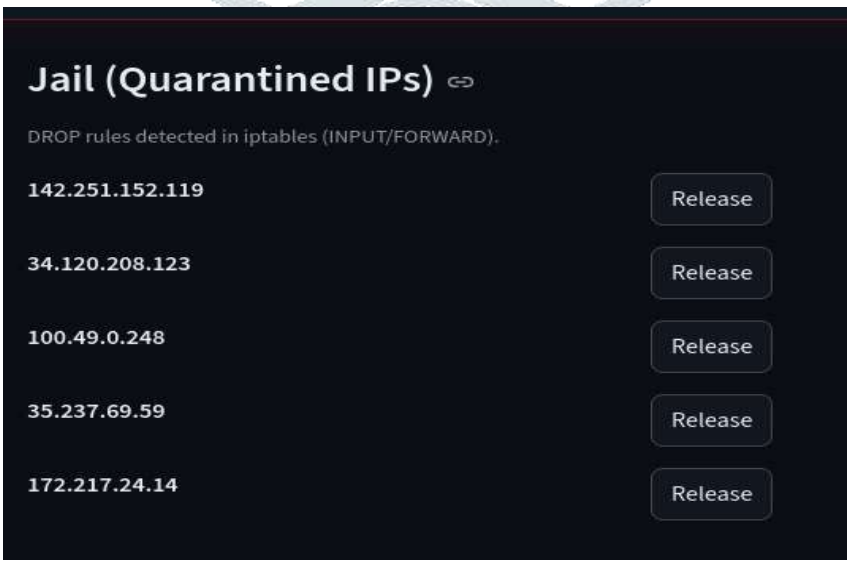
SCREENSHOTS



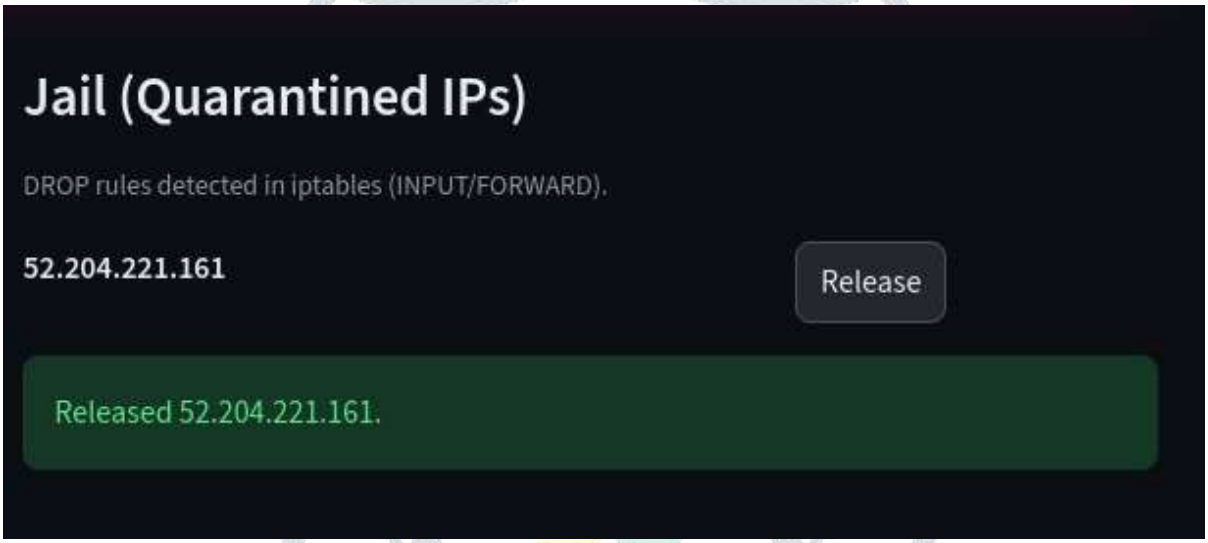


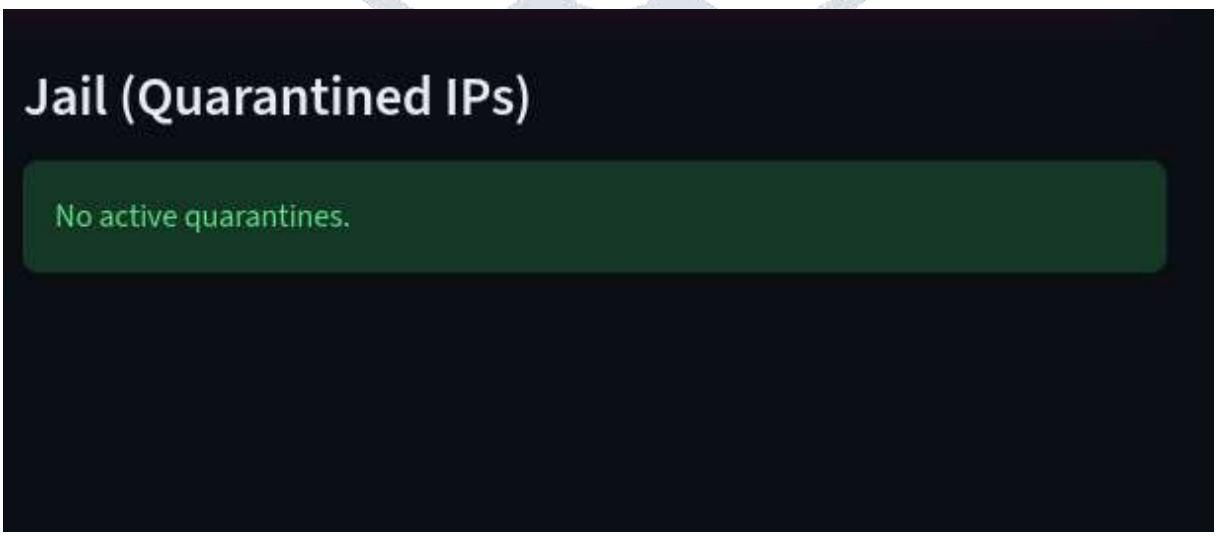


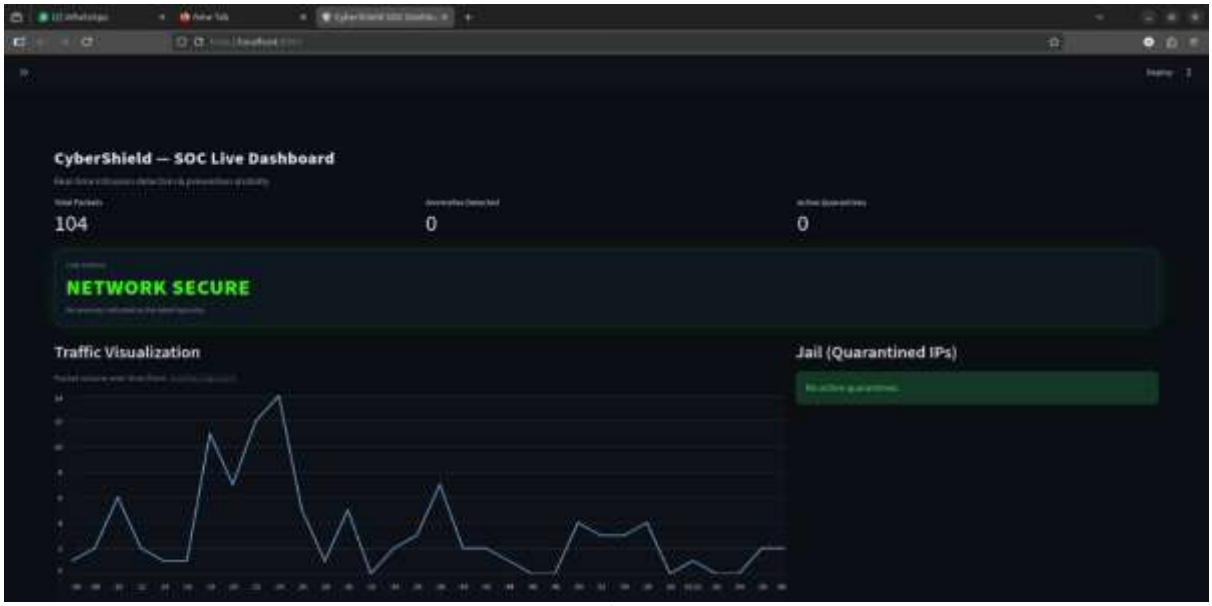



Jail (Quarantined IPs)
DROP rules detected in iptables (INPUT/FORWARD).
142.251.152.119
Release
34.120.208.123
Release
100.49.0.248
Release
35.237.69.59
Release
172.217.24.14
Release


Recent Incidents
ts
reason
flagged_ip
flagged_mac
dns_qname
score
is_malicious
None ml: anomaly detected (IsolationForest) 13.107.5.93 52:55:0a:00:02:02 None -0.0035
None ml: anomaly detected (IsolationForest) 13.248.241.7 52:55:0a:00:02:02 None -0.0032
None ml: anomaly detected (IsolationForest) 13.248.241.7 52:55:0a:00:02:02 None -0.0032
None ml: anomaly detected (IsolationForest) 13.248.241.7 52:55:0a:00:02:02 None -0.0032
None ml: anomaly detected (IsolationForest) 52.3.138.207 52:55:0a:00:02:02 None -0.0081
None ml: anomaly detected (IsolationForest) 52.3.138.207 52:55:0a:00:02:02 None -0.0081
None ml: anomaly detected (IsolationForest) 54.85.4.37 52:55:0a:00:02:02 None -0.0081
None ml: anomaly detected (IsolationForest) 54.85.4.37 52:55:0a:00:02:02 None -0.0081
None ml: anomaly detected (IsolationForest) 54.85.4.37 52:55:0a:00:02:02 None -0.0081
None ml: anomaly detected (IsolationForest) 35.172.207.234 52:55:0a:00:02:02 None -0.0078


Jail (Quarantined IPs)
DROP rules detected in iptables (INPUT/FORWARD).
52.204.221.161
Release
Released 52.204.221.161.


Jail (Quarantined IPs)
No active quarantines.



VI. CONCLUSION

The CyberShield project successfully demonstrates that a robust, real-time Intrusion Prevention System can be implemented using affordable hardware and optimized software architectures. By utilizing the Atheros AR9271 adapter as a gateway and a Python-based engine for traffic scoring, the system overcomes the limitations of passive monitoring. The transition from resource-heavy SQL databases to a lightweight CSV logging system proved to be an efficient design choice, significantly reducing CPU overhead and making the system suitable for edge computing on low-power devices. Ultimately, CyberShield provides a functional "Defensive Gateway" that bridges the gap between sophisticated network forensics and user-friendly, automated network defense.

The success of the CyberShield project underscores a pivotal shift toward decentralized, hardware-integrated security solutions that prioritize efficiency over complexity. By proving that a high-performance IPS can operate effectively using a flat-file CSV architecture and consumer-grade wireless adapters, this research challenges the notion that robust network defence is exclusive to expensive, enterprise-level infrastructures. The project demonstrates that when security tools are stripped of unnecessary database overhead and focused on the "Defensive Gateway" model, they become significantly more agile, allowing for deployment in diverse and resource-constrained environments where traditional systems would fail to operate.

Ultimately, CyberShield serves as a scalable blueprint for the future of democratic network security, empowering individual users and small organizations to take full ownership of their digital perimeters. The integration of real-time visualization through Streamlit with autonomous DNS-based mitigation creates a proactive security posture that reduces the reliance on constant human oversight. As the landscape of cyber threats continues to evolve toward more automated and persistent attacks, the principles demonstrated in this project—simplicity, hardware-level interception, and immediate response—will remain essential components in building a more resilient and secure global network edge.